Band-1 Hadamard Order 7604 — Paley-I IN SPK (q=7603); Miyamoto SKIP (no H_1900); Gap 2052 Still OPEN

H_7604 Paley-I PASS → 7604 IN SPK; Miyamoto SKIP; Paley-II/Kronecker SKIP; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-16

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7508_Miyamoto`; `2026-09-16_Band1_Hadamard_7492_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7604_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims a **finite construction of order 7604 only** (verified $\max|HH^\top - nI| = 0$), via classical Paley-I H_{7604} = Paley-I($q = 7603$) (prime $\equiv 3 \pmod 4$). Classical SPK for 7604: Paley-I **PASS**es ($q = 7603$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL**s ($q = 3801 = 3 \cdot 7 \cdot 181$ NOT prime power); Kronecker **FAIL**s (4×1901, 1901 prime $\equiv 1 \pmod 4$); Miyamoto **SKIP** (no classical Paley/Kronecker seed for H_{1900}). Therefore order **7604 is IN SPK via Paley-I**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7508 Miyamoto+Paley-I IN SPK**. Prior SPK seeds **7508 / 1876 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **7604** folded into ONBOX/SPK when verified. Order **7556** SKIPPED this dash (no H_{472} seed).

Abstract

Finite construction of order **7604**.

- H_{7604} = Paley-I ($q = 7603$) — classical SPK DIRECT
- Paley-II for 7604: **SKIP** ($q = 3801 = 3 \cdot 7 \cdot 181$ NOT prime power)
- Kronecker for 7604: **SKIP** ($4 \cdot 1901$ not admissible)
- Miyamoto for 7604: **SKIP** (no H_{1900} seed via Paley-I/II/Kronecker)

SPK: 7604 IN SPK via Paley-I q=7603. Prior SPK **7508 / 1876 / 1872 / 1860 / 7444 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7604.py`

Results. overall_pass=true; units 104/104; wall 355.7962s; max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7604 = 4 \cdot 1901$; 1901 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible *in principle*, but seed H_{1900} has no classical Paley-I ($q = 1899$ not prime), Paley-II ($q = 949 = 13 \cdot 73$), or Kronecker (4×475) path → Miyamoto **SKIP**.
- SPK: Paley-I $q = 7603$ **PASS**; Paley-II $q = 3801$ **FAIL** (NOT prime power); Kronecker **FAIL → IN SPK via Paley-I**.
- Prior: 7508 Miyamoto+Paley-I IN SPK shipped; 7556 SKIPPED (no H_{472}); this ship $q = 1901 \rightarrow n = 7604$ via Paley-I DIRECT.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_7604 Paley-I	Paley-I q=7603 DIRECT	0	350.7971
H_7604 Paley-II	SKIP q=3801=$3 \cdot 7 \cdot 181$	—	—
H_7604 Kronecker	SKIP $4 \cdot 1901$ not admissible	—	—
H_7604 Miyamoto	SKIP no H_1900 seed	—	—

SPK closure note

Seeds include prior SPK ships through **7508** lineage (**7508** Paley-I; **1876** Paley-II; **1872** Paley-I; **1860** Paley-II; **7444** Paley-II GF(61^2); **1848** Paley-I; **7396** Paley-II; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7492/7156/7108/6836/6788/6676/6484/6404/6388/6196)

+ verified **7604** (Paley-I); **exclude** Miyamoto-only bonuses. Known count 1546. Next constructive gap **2052** (OPEN).

Quantity	Value
7604 in SPK	**True** (Paley-I q=7603)
7508 / 1876 / 1872 / 1860 / 7444 in SPK	**True** (prior)
7492 / 7156 / 7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite construction of 7604 ONLY · SPK via Paley-I only · Miyamoto not attempted (no H_1900).

Repro

Lab: `/workspace/experiments/band1/hadamard_7604_paleyI/`
Artifacts: `H7604_paleyI_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.